

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on

the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted [online](#).
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: Uganda
- Organization(s) or entity(ies) responsible for the preparation of the report*:

Name: Uganda National Commission for UNESCO

Website:

Email address*:

Phone number*:

Please describe the role/mandate of your organization: To liase between UNESCO and the Government of Uganda and individuals, Agencies and Civil Society Organisations as well to contribute to the attainment of peace, justice, respect for human rights, freedom

and security in Uganda through education, sciences, culture, and communication and information.

- Officially designated contact person(s) who completed this survey:

Full name*: Dr Dominic Venture Mundrugo-Ogo Lali

Position*: Deputy Secretary General/Focal Person Natural Sciences

Email address*:

- Full Name(s) of designated official(s) certifying the report: Dr Rosie Agoi
- Brief description of the consultation process established for the preparation of the report:

Summary Description of the Consultation Process

The consultative process involved the members of the Natural Sciences Programme Committee and members of the Specialized National Committees of MAB, IBSP, IHP and IGCCP, all of whom are drawn from the various Ministries, Departments and Agencies whose areas of operation and competence are in the Natural Sciences. A dissemination meeting was held on 17th December 2024 attended by 24 out of the targeted 40 officials who participated both physically and online. During the meeting, the Deputy Secretary General/Focal Person for Natural Sciences and Open Science welcomed everyone to the meeting and informed them that the meetings objectives were to create awareness, understand the principles, build consensus on the way forward, guidelines on submitting reports and encourage collaboration and networking to foster a culture of open science in order to align with advocacy for open science policy. Participants were introduced to the UNESCO OPEN SCIENCE TOOLKIT and the Guidelines for preparation of the report.

The key messages were that communities are interconnected, there is a framework for member states to develop open science and domesticate it and there are ways to promote open science through reading, talking and writing about it. In terms of practice, scientific knowledge should be made accessible and the products be open to everyone. Open science is a contrast to ‘closed’ science. In addition, it calls for engagement and citizen science and volunteering from the community. Open science looks for technologies that make information as available as possible while taking into account copyright infringement and endangered species and communities. Open science is needed for acceleration of the achievement of the SDGs among other things. It would be used through shared understanding of open science, enabling policy environment, investment in infrastructure, training, educating and capacity building, promoting innovative approaches. It is to be used by all research institutes and organizations that practice, regulate and promote science and all researchers. It is a non-binding recommendation however it carries considerable political and moral authority.

The values of Open Science were shared as: equity and fairness, quality, collective benefit, diversity and inclusiveness. Similarly, the principles of open sciences including sustainability, collaboration among others were shared. All were called to join in the implementation of the recommendations of Open Science.

In their feedback, the participants said the following:

- One participant, a University Lecturer, said he had been involved in citizen science to implement a project in Lake Albert region to help the community. He recommended that the dissemination be extended to lower levels like secondary and primary schools
- Another recommended for more time of about a year to be able to report of implementation progress.

- Others noted the need for replication of open science works, transparency of data of sciences and the issue of resources and the roles and responsibilities of the participants in the reporting be clearly specified.
- A participant, who is a librarian by training complained that librarians have not been brought on board to champion open science yet they could be the very ones at the core of the implementation of the recommendation. He noted that only 4 institutions in Uganda have institution repositories and there is need to ensure other universities do the same. It was recommended that libraries be funded to establish institutional repositories and come up within a policy framework that ensures university administrations adhere to.
- The Executive Director of the Association of Vice Chancellors of the Universities in Uganda welcomed the normative instrument on Open Science and strongly recommended the necessity of establishing the necessary infrastructures needed for open science. In his view, the recommendation should be widely disseminated and a policy framework be developed to guide government to facilitate infrastructure for science including the archiving for sciences. He also proposed the need for a National Monitoring Framework. He recommends new stakeholders like the Regional Inter-University Council (IUCEA), ministers and all higher education officials, among others be brought on board.

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Kyambogo university, (Faculty of Science)

Website*: <https://kyu.ac.ug/>

Email address*

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Makerere University, (Col. of Natural sciences: Dept. of Physics, Mathematics, Hydrological Sciences, Mathematics)

Website*: <https://mak.ac.ug/>

Email

addresses:

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: National Environment Management Authority (NEMA)

Website*: https://www.nema.go.ug/new_site/

Email address:

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Uganda National Council for Science & Technology

Website*: www.uncst.go.ug.

Email address:

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Mbarara University of Science and Technology

Website*: www.must.ac.ug

Email address:

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Uganda National Academy of Sciences (UNAS)

Website*: <https://unas.org.ug/>

Email address:

Sector: ☐ Public ☐ Private ☒ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of organization*: International Business, Science and Technology (ISBAT) University,

Website*: <https://www.isbatuniversity.ac.ug/>

Email address:

Sector: ☐ Public ☒ Private ☐ Civil Society ☐ Other

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*: Forum for African Women Educationists (FAWE) UG Alumni Network

Website*: <https://faweuganda.org/>

Email address:

Sector: ☐ Public ☐ Private ☒ Civil Society ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES

If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The Universities namely : Kyambogo university, Makerere University ISBAT University and Mbarara University of Science and Technology. Ministries and Agencies include: Ministry of Water and Environment, Uganda National Council for Science & Technology, National Agricultural Research Organization, Office of the President (Science, Technology and Innovation Secretariat), Ministry of Education and Sports, National Environment Management Authority (NEMA), Ministry of Energy and Mineral development, Petroleum Authority of Uganda, Uganda National Meteorological Authority, Uganda Wildlife Authority, Dept of Geological Mines and Surveys /Geoparks , Uganda National Academy of Sciences (UNAS), Mulago National Referral Hospital, National Forestry Authority, Ministry of Tourism, Wildlife and Antiques, Youth Committee, Atomic Energy Council, Uganda Institution of Professional Engineers, Forum for African Women Educationists (FAWE) UG Alumni Network, FAWE UG Alumni Network, National Water & Sewerage Corporation,

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

NO

1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES

If yes, please provide details on the activities organized or foreseen and links as relevant.

Activities Organised :

- i) Dissemination of the open to scholars and experts, decision makers, private sector and the public at large.
- ii) Engaging the private sector in the discussion about the ways in which the scope of open science principles and priorities can be enlarged and mutually shared.
- iii) Enabling open multi-stakeholder discussions on open science benefits and its real and apparent challenges as regards, for example, competition, extraction and exploitation of data by more advanced technologies, links to intellectual property rights, privacy, security and inequalities between publicly and privately funded research, in order to address these challenges constructively and implement open science practices in line with the values and principles outlined in the Recommendation.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

NO

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

There is need for experts, high-level inter-ministerial involvement and funding which the National Commission may not be able to muster in its current configuration within one ministry.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

YES

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information (**See attached**)

Title of the policy: **National Science, Technology & Innovation Policy.**

The authority in charge of development of the policy: The **Uganda National Council for Science and Technology (UNCST)**, which was established by **Government** in 1990 initially under the Ministry of Finance, Planning and Economic Development, but later under the Office of the President overseen by the Minister for Science, Technology and Innovation.

Entities involved and the process of development,

year of adoption,

year of last update: ?????

key areas/sections of the policy: Background to the STI Policy, Guiding Principles, Goal and Objectives, Policy Statements and Strategic Actions, *Support* for STI Policy Implementation

The Strategic Actions are namely:

- Create an Enabling Policy Environment to Foster STI and Augment their Contribution to National Development
- Build STI Sector Capacity for Generation and Transfer of Technology.
- Establish and Strengthen Legal and Regulatory Framework to Ensure Ethics and Safety in STI Development and Application.
- Strengthen the STI Coordination Framework to Enhance Sector Performance.

links to the webpage:

<https://www.uncst.go.ug/manage/files/downloads/National%20STI%20Policy%202009.pdf>

- does it include any of the following open science elements:
 - open access to scientific knowledge
 - open science infrastructure
 - open engagement of societal actors
 - open dialogue with other knowledge systems

Yes, the above elements exist in the policy although not as Open Science as the concept of Open Science had not been conceived in 2009 when the policy was developed and adopted for use. .

Even now, the Open Science Recommendation has not yet been domesticated in the country.

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

NO

The Open Science Recommendation has not yet been domesticated in the country.

- 2.3 In your country, are there specific policy instruments^v that aim to promote open science?

NO

The Open Science Recommendation has not yet been domesticated in the country. It may need some special expertise to develop the relevant policy instruments.

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

NO

The Open Science Recommendation has not yet been domesticated in the country. It may need some special expertise to develop the relevant policy instruments to operationalise the Open Science Recommendations.

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

NO

- 2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (ref.: [\(ii\) 17.j, \(iii\) 18.i, \(iv\) 19.c](#))

NO

The Open Science Recommendation is yet to be domesticated and operationalised in the country.

- 2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

NO

The Open Science Recommendation is yet to be domesticated and operationalised in the country.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

NO

However, we will try to work towards realising the targets in (ii) 17i.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.j, 23.c](#))

NO

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

The Gross domestic expenditure on R&D as a share of GDP in Uganda in the period 2020-2022 amounted to 0.14 percent. R&D spending as a percentage of the GDP has remained constant since 2020. In absolute terms, the country's gross spending on R&D amounted to 150 million U.S. dollars in 2022 ([Saifaddin Galal](#), Sep 29, 2023, <https://www.statista.com/statistics/1345231/gerd-as-gdp-share-in-uganda/#:~:text=In%202022%2C%20the%20gross%20domestic,GDP%20amounted%20to%200.14%20percent>).

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

There were **13.30 million** internet users in Uganda in January 2024. This translates to **27.0 percent** of the total population that has access to the internet at the start of 2024 meaning that that **35.95 million** people in Uganda which is **73.0 percent** of the population remained offline at the beginning of the year 2024(<https://datareportal.com/reports/digital-2024-uganda/#:~:text=Uganda's%20internet%20penetration%20rate%20stood,January%202023%20and%20January%202024>). According to the Permanent Secretary, Ministry of Information Communication and Technology (ICT), Internet users in Uganda showed steady increase from year to year, For example, internet users increased by 1.2 million (10.3%) between January 2023 and January 2024 pushing the total number of internet users in Uganda to 13.30 million(27.0% of total population by Mar 2024.

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\) 18.c](#))

YES

There is a Research and Education Network for Uganda (RENU), a not-for-profit National Research and Education Network (NREN) that promotes collaboration among its member institutions and their global partners. The network is a cooperatively owned and community-driven service provider that helps to facilitate research and education networking among its member institutions through interconnecting them with other research and education networks as well as to the commercial Internet worldwide, thereby overcoming the traditionally high costs of information and knowledge sharing and exchange. ([https://www.kab.ac.ug/partner/research-and-education-network-for-uganda-renu/#:~:text=Research%20and%20Education%20Network%20for%20Uganda%20\(RENU\)%20is%20a%20not,institutions%20and%20their%20global%20partners](https://www.kab.ac.ug/partner/research-and-education-network-for-uganda-renu/#:~:text=Research%20and%20Education%20Network%20for%20Uganda%20(RENU)%20is%20a%20not,institutions%20and%20their%20global%20partners)) The website is: <https://renu.ac.ug/>

There is a National Research Information Management System (NRIMS) that is an integrated online platform designed to serve researchers and research users (<https://nrims.uncst.go.ug/index.php?s%C3%BCperbetin+giri%C5%9F+sayfas%C4%B1+>)

- 3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (ref.: [\(iii\) 18.d, 18.e, 18.k, 18.l, 18.m](#))

UNDER DEVELOPMENT

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\) 18.e, 18.f](#))

NO

Open science infrastructures at regional and international levels not yet in place.

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

NO.

4. **INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE**

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

NO

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

NO

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

NO

All bice Chancellors of Universities resolved to start operationalisation of the Recommendation

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

NO

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

NO

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20i](#))

YES

All the vice chancellors in their meeting in 2024 adopted resolution

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

NO

This area is yet work in progress.

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

NO

No plans yet.

- 5.4 Have there been any unintended negative consequences^{xi} of open science practices reported in your country? (ref.: [\(v\) 20](#))

NO

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

√YES, at the national level.

√YES, at the institutional levelsl.

√YES, at both national and institutional levels.

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- √promoting open innovation practices.
- √Others (with a box to add text)

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a\)](#)

Wider dissemination is being conducted on the Recommendation on Open Science. This is to be followed by the following actions:

- (i) *Encouraging international scientific collaborations, as one of the integral practices of open science and the most important driving factor for an intensive exchange of scientific knowledge and experience, as well as the paramount for the openness of science.*
- (ii) *Supporting the creation and maintenance of effective collaborative networks to exchange best open science practices and lessons learned from the design, development and implementation of open science policies, initiatives and practices.*

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b, 22.e, 22.f\)](#)

NO

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=21>)

NO

8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

NO

i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).

ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

-
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
 - iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
 - v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
 - vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
 - vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
 - viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
 - ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
 - x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
 - xi For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.